

AUSTIN BRODERICK MCELROY

Embedded Software Engineer | Rust Firmware | Medical Devices

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EXPERIENCE

Senior Staff Software Engineer - Pictorus / Renesas  10/2024 – Present  Remote – Austin, Texas

- Designed, assembled, and brought up an STM32G474 hardware-in-the-loop test PCB covering UART, I2C, SPI, GPIO, ADC, DAC, CAN, and PWM; integrated it into automated nightly end-to-end testing.
- Led migration of an embedded linear algebra library from heap allocation to static allocation, eliminating allocator dependence and fragmentation risk in long-running `no_std` firmware. Open sourced as [pictorus-rs](#).
- Built and demonstrated a custom `no_std` Rust HAL for an unsupported Renesas MCU in two weeks, covering power, I2C, GPIO, and timers. Open sourced as [ra4m2-hal](#).
- Integrated Rust firmware into customer C and Rust environments, including NuttX/PX4 and FreeRTOS, using static-library interfaces and hands-on hardware debugging.
- Owned a cross-platform CLI for telemetry and OpenOCD firmware updates; improved telemetry bandwidth 4x by replacing JSON with Postcard serialization.
- Led delivery of protocol drivers and code generation for PWM, ADC, DAC, and SPI across STM32, Renesas, and embedded Linux targets in Rust, Python, and TypeScript.

Staff Scientist - BridgeSource Medical  08/2020 – 10/2024  Austin, Texas

- Architected firmware and telemetry software for four medical-device SBIR projects totaling \$8M, using embedded Rust, bare-metal C, and Zephyr RTOS across Cortex-M microcontrollers.
- Implemented real-time acquisition, output, and signal processing up to 1 MHz using DMA, interrupts, timers, and custom SPI, I2C, UART, PWM, GPIO, and CAN interfaces.
- Led a \$250k breath-adaptive nebulizer program through successful Phase I human testing, helping secure \$2.5M in Phase II funding; designed its two-stage boost and three-channel piezo drive electronics.
- Built `no_std` Rust/Embassy firmware for an nRF52840 and a Flutter iOS app over BLE to control haptic shoe orthotics used in a small clinical trial.
- Developed bare-metal TM4C and Zephyr/nRF52840 firmware with inter-processor communication, BLE device firmware updates, and LVGL-guided device workflows.

CEO / Engineering Consultant - RaptorView, LLC

 01/2010 – Present  Austin, Texas

- Designed and brought up a custom ESP32-S3 PCB for a 64x64 RGB [Game of Life matrix](#), including USB-C PD, a 27W buck supply, HUB75 level shifting, and native USB/JTAG; wrote Rust firmware on ESP-IDF and a reusable `no_std` simulation core.
- Developed and released [Historicle.ai](#), a Rust journaling app with local LLM support and extensive automated Rust and front-end test suites. [Available in the macOS App Store](#).
- Designed and fabricated a USB-C powered LED drive circuit, Rust firmware (nRF52840), and React Native iOS app for a BLE controlled, [LED light engine](#).
- Architected multi-threaded C# data acquisition (DAQ) and display GUI software for a vaginal OCT system and supported the system through clinical trial.
- Designed, simulated, laid out, and manufactured a high-voltage flash-lamp circuit for acne treatment; managed the successful FDA 510(k) application (K111937).
- Architected, released, and open sourced a package for OpenCL in LabVIEW for high speed computing. [OpenCLV](#)

ADDITIONAL SKILLS

Embedded Linux

Unit Testing

Integration Testing

Git

CMake

Yocto

KiCad

SPICE

Docker

Schematics

Datasheets

Technical Documentation

Troubleshooting

EDUCATION

M.Sc.E.E, University of Texas, Austin

B.Sc.E.E, University of Texas, Austin

TECHNICAL SKILLS

Language	Frameworks / Tools
Rust	no_std, Embassy, ESP-IDF, RTIC
C	Bare metal, Zephyr, FreeRTOS, LVGL
Python	FastAPI, TensorFlow, NumPy
TypeScript	React Native, Svelte, Vue
LabVIEW	DAQmx, C#/C++ DLLs, FPGA

EMBEDDED SKILLS

HIL Testing

JTAG

OpenOCD

UART

I2C

SPI

DMA

PWM

ADC

DAC

CAN

BLE

GPIO

Timers

Interrupts

HAL

Cortex-M

ESP32-S3

STM32

nRF52840

TM4C123

Logic Analyzer

Oscilloscope

Buck/Boost

Analog Filters

Board Bring-up

Hardware Debugging

Device Drivers

ADDITIONAL EXPERIENCE

Research Scientist - University of Texas – Austin 📅 06/2012 – 05/2020 📍 Austin, Texas

- Designed PCB hardware and wrote firmware for the [UT ABBU project](#), an emergency breathing-assist system developed for hospitals during COVID-19.
- Developed a TensorFlow semantic-segmentation pipeline for OCT using U-Net and autoencoder architectures, resulting in two patents.
- Accelerated OCT phase processing for blood-vessel detection 600x with OpenCL.
- Designed and built custom 400 MHz photodetectors for high-power near-infrared laser systems.
- Led LabVIEW DAQ and OpenCL image-processing development across five OCT platforms, supporting dozens of publications.
- Architected software for the LAMPS 3D printer, integrating FPGA and microcontroller subsystems with high-speed thermal imaging and helping secure Phase II funding.

R&D Engineer - Volcano Corporation 📅 06/2006 – 01/2010 📍 Austin, Texas

- Pioneered GPU computing and optimal hardware and software methods for non-linear re-sampling in real time for OCT.
- Designed differential analog photodetector circuits and filters for OCT frontends up to 100MHz, saving over \$100k.

SELECT PATENTS (3 OF 6)

- Milner, Thomas E, Austin McELROY, Aydin Zahedivash, and Nitesh Katta (Oct. 2023). *Multi-channel orthogonal convolutional neural networks*. US Patent 11,779,220.
- Kemp, Nathaniel J, Austin Broderick McElroy, and Joseph P Piercy (May 2016). *Real time SD-OCT with distributed acquisition and processing*. US Patent 9,347,765.
- Kemp, Nathaniel J, Roman Kuranov, Austin Broderick McElroy, and Thomas E Milner (Nov. 2011). *Apparatus and methods for uniform frequency sample clocking*. US Patent 8,049,900.

SELECT PUBLICATIONS (4 OF 72)

- Katta, Nitesh, Arnold D Estrada, Austin B McElroy, Aleksandra Gruslova, Meagan Oglesby, Andrew G Cabe, Marc D Feldman, RY Declan Fleming, Andrew J Brenner, and Thomas E Milner (2019). "Laser brain cancer surgery in a xenograft model guided by optical coherence tomography". In: *Theranostics* 9.12, p. 3555.
- Katta, Nitesh, Austin B McElroy, Arnold D Estrada, and Thomas E Milner (2018). "Optical coherence tomography image-guided smart laser knife for surgery". In: *Lasers in surgery and medicine* 50.3, pp. 202–212.
- Wang, Tianyi, Austin McElroy, David Halaney, Deborah Vela, Edmund Fung, Shafat Hossain, Jennifer Phipps, Bingqing Wang, Biwei Yin, Marc D Feldman, et al. (2015). "Dual-modality fiber-based OCT-TPL imaging system for simultaneous microstructural and molecular analysis of atherosclerotic plaques". In: *Biomedical optics express* 6.5, pp. 1665–1678.
- Kuranov, Roman V, Austin B McElroy, Nate Kemp, Stepan Baranov, Joe Taber, Marc D Feldman, and Thomas E Milner (2010). "Gas-cell referenced swept source phase sensitive optical coherence tomography". In: *IEEE Photonics Technology Letters* 22.20, pp. 1524–1526.

SELECTED EMBEDDED SYSTEMS

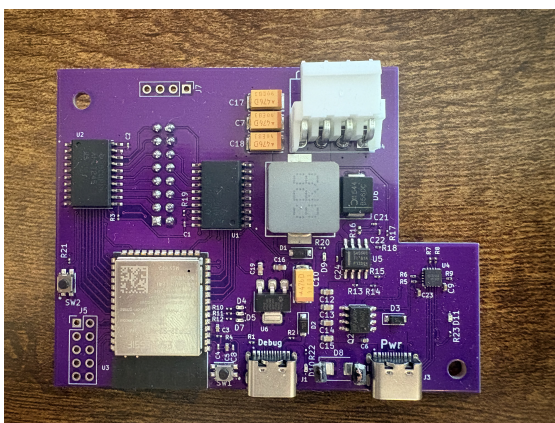


Figure 1: Custom ESP32-S3 Game of Life PCD with USB-C PD, 20 W buck, HUB75, native USB/JTAG, WiFi, and Rust on ESP-IDF/FreeRTOS.

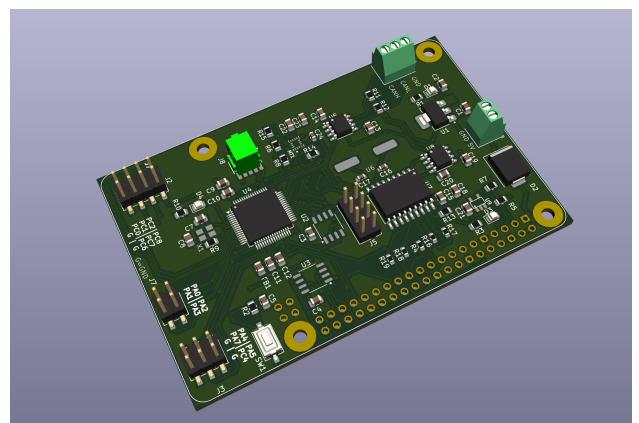


Figure 2: STM32G474 HIL PCB with IMU, CAN, JTAG, FRAM, mixed-signal I/O, and automated nightly protocol tests.